SHASU VATHANAN

SHASU VATHANAN - GEN AI - PRODUCT MANAGER

GEN AI ARCHITECT JOURNEY · PHASE-004

Student Grade App

Project brief and build specification

Goal	Turn a mark into a letter grade in the browser
Difficulty	Beginner to Medium
Tools	Python · Streamlit
Single file	grade_app.py
Phase	Phase-004 - Interfaces and APIs
Folder	Phase-004-FastAPI-And-Streamlit

Shasu Vathanan - GEN AI - Product Manager

SHASUVATHANAN.COM · linkedin.com/in/shasuvathanan

1. What you will build

A small web app using the Streamlit framework that turns a mark into a letter grade.

Instead of running in the terminal, your app opens in the browser with a simple user interface: the user types or selects a mark - a number between 0 and 100 - in an input widget, and the app displays the matching grade on the page. You will set up a clean Python environment, install Streamlit, write your code in a **.py** file, run the app, and confirm it shows the correct grade.

Difficulty	Beginner to Medium
Tools	Python, Streamlit

2. Grading scale to implement

Your app must convert a mark into a grade using exactly this scale:

Mark range	Grade
90 - 100	A
80 - 89	B
70 - 79	C
60 - 69	D
Below 60	E

Inclusive boundaries

A mark of exactly **90** is an A, exactly **80** is a B, exactly **70** is a C, and exactly **60** is a D.

3. Requirements

- Use the **Streamlit** framework to build the user interface. Pure Python for the grading logic.
- Provide an input widget where the user can enter or choose a mark.
- Display the entered mark and the resulting grade clearly on the page.
- Handle the full **0-100** range and cover every grade band above.
- Give the app a title or heading so it looks finished.
- Keep all your code in a single file named **grade_app.py**.

Think about edge cases

What should happen if someone enters a number above 100, below 0, or leaves the field empty? Decide how your app should respond, and make sure it shows a friendly message instead of crashing.

4. Step-by-step setup

Follow these steps in order. Commands are shown for both macOS/Linux and Windows where they differ.

Step 1 - Create a project folder

SHELL
```
mkdir grade-app
cd grade-app
```

Step 2 - Create a virtual environment (venv)

A virtual environment keeps this project isolated from the rest of your system.

SHELL
```
python -m venv venv
# (on some systems use 'python3' instead of 'python')
```

Step 3 - Activate the virtual environment

SHELL
```
# macOS / Linux
source venv/bin/activate

# Windows (PowerShell)
venv\Scripts\Activate.ps1

# Windows (Command Prompt)
venv\Scripts\activate.bat
```

When activated, you will see **(venv)** at the start of your terminal prompt.

Step 4 - Install Streamlit

Install the Streamlit framework inside the activated venv.

SHELL
```
pip install streamlit
```

Step 5 - Run your app

Streamlit apps are started with **streamlit run**, not **python**.

SHELL
```
streamlit run file_name.py
```

The app opens automatically in your browser at **http://localhost:8501**. Test it with several marks to confirm each grade band works.

Step 6 - Stop and deactivate when finished

Press **Ctrl + C** in the terminal to stop the app, then deactivate the venv.

SHELL
```
deactivate
```

5. How the app should behave

Your layout and wording can differ, but a typical interaction should look like this:

TEXT
```
[ Enter your mark (0-100): 85 ]

Mark: 85 -> Grade: B
```

6. Deliverables

- All application code and grading logic in a single file named **grade_app.py**.
- A **requirements.txt** so the project can be recreated on any machine.
- Screenshots of the running app for at least three different marks, covering more than one grade band.
- A **README.md** that documents the grading scale and explains how invalid input is handled.
- The whole project committed to GitHub.

Definition of done

Every grade band returns the correct letter, both inclusive boundaries and decimal marks behave correctly, and empty, non-numeric, and out-of-range input each produce a readable message instead of a crash.

Next

The completed build for this brief is in this folder: see **grade_app.py**, **screenshots/**, and **README.md**.

Shasu Vathanan - GEN AI - Product Manager

SHASUVATHANAN.COM · linkedin.com/in/shasuvathanan